# Exit Ticket · reflect and retrieve

## Exit Ticket

A branch is always impossible when its subtotal exceeds the target, even if negatives remain.

## Use key vocabulary from memory

1. Explain backtracking in your own words.

2. Explain search tree in your own words.

3. Explain pruning in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Lesson 068 · Backtracking: Choices, Dead Ends and Undo

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Describe the divide-and-conquer approach to solving a problem.

### Q2

What is a dead end in a search?

### Q3

Why must recursion have a base case?

### Q4

Stack Push 2, Push 5, Pop, then Push 8: give popped item and remaining stack.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Define and predict — Backtracking explores a choice, tests whether its state remains viable, then undoes the choice if it cannot lead to a solution.

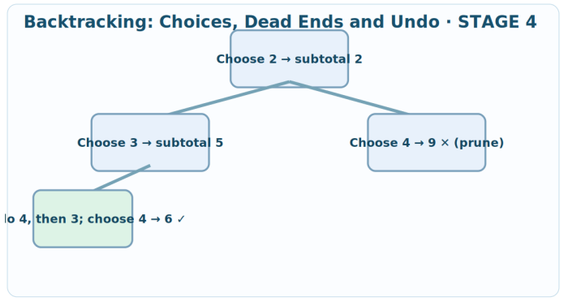
## Core idea 2

Explain — A search tree records choices and alternative branches. Pruning avoids expanding branches already proven invalid.

## Core idea 3

Evaluate — A correct search needs a stopping condition. Pruning on sum>target only works when remaining numbers cannot reduce the sum (e.g. non-negative inputs).

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Explain why pruning when sum>target may be invalid if negative integers occur; give counterexample. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Trace first failed and one successful branch for subset sum target6 from [2,3,4]. [6]

## Question 2

Explain a recursive backtracking design with choose/exclude and two stopping conditions. [6]

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

backtracking — Try and undo choices.

search tree — Branches of possible decisions.

pruning — Stop provably impossible branch.

state — Current partial solution data.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

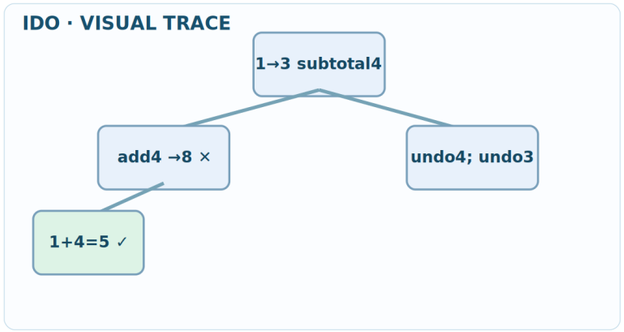
## Problem

Find a subset of [1,3,4] with total 5.

## Follow the worked method

1. Follow the reasoning — Return from the failed choice of4 before excluding3. The state returns to [1], then choosing4 reaches5. Recursive local totals restore on return; mutable state needs an explicit undo.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Follow the reasoning
The alternative branch 2+7 gives a valid solution.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.